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SICK
Sensor Intelligence.

Machine Safeguarding Evaluation

ADVANCED



Customer: **Karthick Engineering & Maintenance Consultants (KE**
No 22/50, Manickavasagam street, Kumaran Nagar,
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600050 Chennai
Tamilnadu
India

Customer no.:

Report no.:

Report version:

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1 Project information

1.1 Contacts

Customer contact information		
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1.2 Version control

Version	Date	Description	Editor
	12/31/2026	Risk Assessment On site	Mr Muthu Vignesh R
	2/16/2026	Second day Visit On site	Mr Muthu Vignesh R

2 Introduction

2.1 Scope of service

- (1) The Machine Safeguarding Evaluation ADVANCED comprises a basic analysis of the safety-related status of one or more machines/ systems or machine or system components. Only obviously identifiable safety gaps that require improvement of existing safety measures or supplementation by further protective measures are considered and analyzed. The results are made available to the client in the form of a report after a 4-eye check, either in printed or electronic form (pdf).
- (2) The service includes the identification of obvious electrical and/ or mechanical hazards based on the "limits of the machine" (intended use and foreseeable misuse). The assessment extends to all operating modes and machine(parts) agreed with the client, after which all risks resulting from the hazards, that are directly or indirectly connected with the machine, are evaluated by a SICK specialist. The assessments carried out are translated into a performance level required (PLr) based on the risk level determined.
- (3) Explanations, descriptions and information from operators, maintenance personnel and management are the primary elements for obtaining information and determining the minimum requirements for risk reduction measures for each machine. The overall effectiveness of the solutions recommended in this report is essentially based on the accuracy and completeness of the information provided by the above individuals.
- (4) Based on the findings obtained, SICK recommends risk reduction measures in order to create the prerequisites for the safe operation of a machine. Recommended measures for risk reduction are based on currently relevant legal requirements, guidelines, requirements from standards on the state of the art and company-specific requirements (if applicable and known to SICK).
- (5) Hazard types that are not of an electrical or mechanical nature (e.g. hazards due to heat, noise, chemicals, radiation, etc.) are identified and documented if they are obvious. SICK will not carry out an evaluation or recommendation of risk mitigation measures for these hazards. For their further evaluation and the respective protective measures, SICK recommends consulting a specialist recognized for this purpose.
- (6) The Machine Safeguarding Evaluation ADVANCED also includes the assessment of existing protective measures for obviously recognizable dirtiness, defects and manipulation.
- (7) The risk assessment process is designed to determine the level of an existing risk in order to make well founded decisions for risk mitigation. In some cases, a residual risk cannot be avoided, so that additional administrative measures are necessary on the part of the customer (e.g. work instructions, signage, use of personal protective equipment, etc.).
- (8) Recommended measures by SICK are based on extensive experience and the interpretation of the applicable standards. The aim of this report is to give an overview of risks at a machine which are hazardous to health in order to develop a suitable strategy for improving machinery safety. With the information of the report, the customer can choose appropriate protective measures while determining the acceptable of the residual risk on a machine. Recommendations for protective measures are always product-neutral.

Machine Safeguarding Evaluation ADVANCED

The following machines were evaluated as part of the machine safeguarding evaluation service. The findings and recommended risk reduction improvements follow for each machine.

Machine type	Manufacturer	Machine name	Asset No.	On-site date
Assembly Machine	Karthick Engineering & Maintenance Consultants (KEMC)	Tilting Machine		2/16/2026

2.2 Disclaimer

- (1) The results of the service "Machine Safeguarding Evaluation ADVANCED" (hereinafter referred to as "the evaluation") are based exclusively on the function of the tested device and its intended use as notified to us at the time of testing by our specialists.
- (2) This report is for information purposes only and does not claim to be complete with regard to all hazards present or occurring on the machine.
- (3) The assessment does not replace a risk assessment in accordance with the Machinery Directive or with the Work Equipment Directive or any other binding requirement which the Client might have to comply with or owe to his employees, representatives, visitors or other persons under contract or law.
- (4) The evaluation considers exclusively electrical and mechanical hazards within the contractual scope of the services (see Section "**Scope of Service**"). Further hazards, risks or circumstances are not subjects of the examination by SICK, this report or the evaluation (hereinafter referred to as "Excluded Circumstances"). Excluded circumstances include, but are not limited to, the following:
 - whether safety functions in their entirety (input - logic - actuators) are implemented in accordance with applicable guidelines and whether required performance levels (PLr) are achieved.
 - hazards that are not of electrical or mechanical nature (e.g. heat, noise, chemicals, radiation, etc.). Such hazards must be assessed separately by the machine operator and, if necessary, dealt with. SICK will endeavor to point out these risks if they are identified by SICK. Risk reduction measures will not be recommended by SICK for such hazards.
 - Elements that could not be identified during testing by our specialists (e.g. changes which are made after the evaluation, etc.).
 - Information which has not been provided, not completely provided – even though SICK has asked for such information in an onsite survey – or mentioned as not applicable or which are misleading (e.g. setups, operating procedures or operating modes which deviate from the agreed processes and procedures examined, etc.).
- (5) SICK SHALL NOT BE LIABLE FOR ANY DAMAGES ARISING OUT OF CIRCUMSTANCES WHICH HAVE NOT BEEN EXAMINED. THE CUSTOMER INDEMNIFIES SICK AGAINST ALL CLAIMS MADE BY THIRD PARTIES AGAINST SICK OR ITS AFFILIATES AS A RESULT OF UNINVESTIGATED CIRCUMSTANCES.
- (6) SICK endeavors to make recommendations that reduce the risk. In some cases, a residual risk cannot be avoided or cannot be avoided with economically justifiable effort, so that additional administrative measures may be necessary on the part of the customer (e.g. work instructions, training, signage, use of personal protective equipment, etc.).

SICK DOES NOT WARRANT THAT THE TESTED MACHINE(S) ARE SAFE AFTER RECOMMENDATION OF RECOMMENDED MEASURES FOR RISK COMMUNISATION.

3 Relevant Regulations and Standards

Within our scope of work and technical expertise, this assessment focuses on risks associated with machine safeguarding hazards, typically associated with moving parts of the equipment or process. The following list includes regulations and standards that were referenced during this assessment within our scope of work.

List of laws and regulations

Name	Date of Issue	Title
2006/42/EG	2006	Machinery Directive

List of standards

Name	Date of Issue	Title
EN ISO 12100	2010	Safety of machinery – General principles for design – Risk assessment and risk reduction
EN ISO 13849-1	2023	Safety of machinery - Safety-related parts of control systems - Part 1: General principles for design
EN 60204-1	2018	Safety of machinery – Electrical equipment of machines – Part 1: General requirements
EN ISO 13850	2015	Safety of machinery - Emergency stop function – Principles for design
EN ISO 13854	2019	Safety of machinery - Minimum gaps to avoid crushing of parts of the human body
EN ISO 13855	2010	Safety of machinery – Positioning of safeguards with respect to the approach speeds of parts of the human body
EN ISO 13857	2019	Safety of machinery - Safety distances to prevent hazard zones being reached by upper and lower limbs
EN ISO 14118	2018	Safety of machinery - Prevention of unexpected start-up
EN ISO 14119	2013	Safety of machinery - Interlocking devices associated with guards - Principles for design and selection
EN ISO 14120	2015	Safety of machinery - Guards – General requirements for the design and construction of fixed and movable guards
EN ISO 4414	2010	Pneumatic fluid power - General rules and safety requirements for systems and their components

4 Machine: Tilting Machine

4.1 Machine identification

Machine data				
Machine name		Tilting Machine		
Manufacturer		Karthick Engineering & Maintenance Consultants (KEMC)		
Machine type		Assembly Machine		
Manufacturing date		2/16/2026		
Re-manufacturing date		2/16/2026		
Model No.				
Serial No.				
Asset No.				
Asset location	Line	No 22/50,		
	Building	Manickavasagam street,		
	Factory	Kumaran Nagar,		
	Street	--		
	City	Chennai	Zip code	600050
	State	Tamilnadu		
	Country	India		

4.2 Relevant documents

The following documents were available during the service.

Documents	Version	Date
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4.3 Limits of the Machine

4.3.1 Use limits

Limits of the Machine						
Intended use	The tilting machine assembly is intended for safe loading, positioning, clamping, and tilting of assembled clutch plates to enable accurate and secure nut installation. It is designed to improve ergonomics, ensure consistent fastening quality, and support controlled unloading after completion of the assembly operation.					
Reasonably foreseeable misuse	Foreseeable misuse includes loading components beyond specified weight limits, bypassing clamping or safety interlocks, operating the tilting function while the component is unsecured, manual handling without the crane, or performing maintenance while the machine is energized, leading to potential injury or equipment damage.					
Expected/ known malfunctions, errors & faults	Known malfunctions may include failure of clamping cylinders, loss of pneumatic or hydraulic pressure, tilting mechanism misalignment, sensor or limit switch failure, control system faults, or crane support malfunction, which could result in improper positioning, unstable loads, or interrupted assembly operations.					
Time limits	Days / year	230	Hours / day	22	Shifts / day	3
Space limits	Fixed workstation with defined loading/unloading zone Clearance for tilting movement and crane operation Restricted access to moving and clamping areas					
Environmental conditions	Indoor industrial environment Normal ambient temperature and humidity Free from excessive dust, oil mist, and corrosive vapors					

Machine parameters		
Processed workpieces/ raw materials		Assembled clutch plate
Capacity		One clutch plate per cycle
Manufacturing process description		The process starts by lifting the assembled clutch plate with a small crane and placing it on the tilting station. The machine clamps the part firmly and tilts it to a comfortable angle. The operator tightens the nuts. After the work is done, the machine returns to position, releases the clamps, and the part is unloaded safely.
Operating modes		Manual Automatic
Power supply	Electrical	Standard industrial electrical supply - 230 V - Single Phase - 50 HZ
	Pneumatic	5 Bar
	Hydraulic	NA
	Other	NA

4.3.2 Machine pictures

Picture(s)	
	Machine Front View
	Job Loading and unloading Tilting Mechanism
	Sick - Safety Rated Controller - SRPCS

4.3.3 Tasks performed at the machine

Affected persons	Skill	Skill description
Operator	Instructed	
Maintenance person	Skilled	
Visitor	Unskilled	

Icon	Task	Affected persons
	Production Task Operator loads clutch plate using crane then places the clutch plate in the machine and starts the process operation with help of cycle start- two hand push buttons	Operator
	Trouble Shooting Identify jam or misalignment Check sensors and actuators Manual repositioning which also covers Preventive Maintenance planned once in 15 days.	Maintenance person
	Cleaning/Maintenance Cleaning chips and oil Lubrication Cylinder or fixture maintenance	Operator
	Visitor Passing by machine Supervising Material movement nearby Persons being present in a hazardous area at or near the machine without a dedicated task at the machine / not working at the machine. This may include just walking by the machine, or observing the operation process without a defined/described associated task.	Visitor
	Gripping Jaw setup Setting process changeover for 280 Model adjustment is done by changing the 3mm bolt and nuts at each stage the HMI has information for the operator for any sensor mis alignment during this process LOTO and administrative controls are followed.	Operator
	Fixture base plate Change Over Setting for new base plate change over for 3 models - 230 , 280 , 310 three different base plates are added in the machine and the change over is done with a frequency (1hr to 1 week) depending on the production demand during this process LOTO and administrative controls are in place	Operator

4.4 Risk summary

Icon	Hazard source	Consequence	Task	Risk ID	Risk IN ⁽¹⁾	Measure advised	Risk OUT ⁽²⁾
	Tilting Movement Motor	Shearing	Production Task	R001	3	Yes	<1
	Pneumatic cylinders	Injection	Production Task	R002	3	Yes	3
	Electrical Power	Electrocution	Trouble Shooting	R003	10	Yes	<1
	Tilting Movement Motor	Shearing	Cleaning/Maintenance	R004	3	Yes	<1
	Short circuit in Control panel	Shock	Trouble Shooting	R005	10	Yes	<1
	Tilting Movement Motor	Shearing	Visitor	R006	3	No	<1
	Pneumatic cylinders	Injection	Gripping Jaw setup	R007	3	Yes	3
	Pneumatic cylinders	Injection	Fixture base plate Change Over	R008	3	Yes	<1

⁽¹⁾ Risk IN: Calculated without considering protective measures

⁽²⁾ Risk OUT: Achievable Risk OUT, if all advised measures have been implemented

Range	Periodization of urgency to modify, improve or inspect risk reduction measures.
AR / <1	Priority 5 (reasonable)
1	Priority 4 (appropriate)
2-3	Priority 3 (urgent)
4-7	Priority 2 (very urgent)
8-10	Priority 1 (immediately)

	The priority statements do not imply that any of the recommendations can be neglected or avoided. It is highly recommended to apply all recommendations in an appropriate time.
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4.5 Machine layouts

4.5.1 Layout current state

Front View



Production Task



Griping Jaw setup



Trouble Shooting



Cleaning/Maintenance



Fixture base plate Change Over



Fix Guard ✓



Warning Signs ✓



Operator Access ✓



Cycle Start PB ✗



Gloves ✗



Safety Shoes ✓



Front Estop ✓



ON/Off Switch ✓



Air isolation ✓



Visitor



Tilting Movement Motor



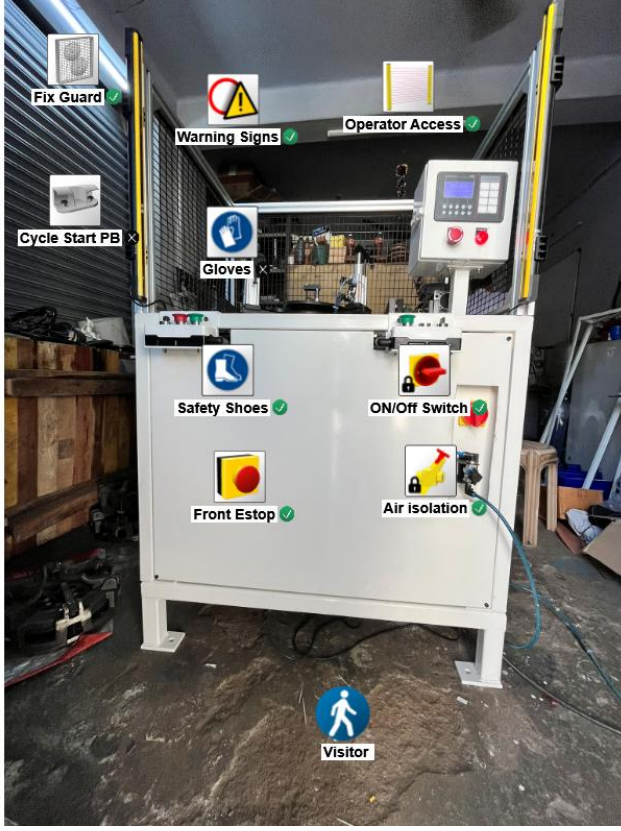
Electrical Power



Pneumatic cylinders



Short circuit in Control panel



4.5.2 Layout future state

4.6 Risk reduction measures

4.6.1 Inherent safe design

4.6.2 Technical protective measures / Engineering controls – Guards

	Fix Guard 	Fixed guard- Machine perimeter	
		Designed and positioned according to ISO 14120 and ISO 13857. For existing measures: Height: 700 mm Length: 900 mm Mesh gap dimensions: 20 mm Make of the guard: Fabricated Typical residual risk: - Failure to replace guard following maintenance, repair or setup Assigned Safety Function(s): SF00 - Permanently preventing entry/access	

4.6.3 Technical protective measures / Engineering controls – Devices

	Air isolation 	Energy isolation pneumatic	PLr c
		Implemented according to ISO 4414. Typical residual risk: - Failure to follow documented procedures - Operation for purposes other than emergency - Device not readily accessible Assigned Safety Function(s): SF11 - Initiating stop and preventing start	
	Cycle Start PB 	Two-hand actuating control	PLr c
		Cycle start system implemented and positioned according to ISO 13851 and ISO 13855. Typical residual risk: - Improper location of device relative to hazard zone(s) - Inappropriate modification of equipment/fastening solution - Control of hazard by person other than exposed operator Assigned Safety Function(s): SFXX - Two Hand Cycle Start Action to be taken	
	Front Estop 	E-Stop push button	PLr c
		To be implemented according to ISO 13850 and IEC 60204-1. Typical residual risk: - Operation for purposes other than emergency - Device not readily accessible - Introduction of obstruction preventing actuation Assigned Safety Function(s): SF99 - Emergency stop	
	ON/Off Switch 	Power Isolation switch	
		Implemented according to IEC 60204-1. Typical residual risk: - Failure to follow documented procedures - Operation for purposes other than emergency - Device not readily accessible Assigned Safety Function(s): SF99 - Emergency stop	
	Operator Access 	Light curtain	PLr c



Type 4 safety rated Light curtain positioned according to ISO 13855.

For existing measures:

Height of top beam from ground: 1400 mm

Height of lowest beam from ground: 900 mm

Resolution: 14 mm

Mounting orientation: Vertical

Distance to hazard: mm

Typical residual risk:

- Inappropriate modification of equipment/fastening solution
- Improper location of detection zone relative to hazard zone(s)

Assigned Safety Function(s):

SF13 - Safeguarding work areas shared by humans and machines

4.6.4 Administrative measures

	A005		Training	
		Workers to be provided adequate training in accordance to the requirements of Use of Work Equipment Directive 2009/104/EC. Workers to be provided adequate training in accordance to the local/regional requirements. Typical residual risk: - Failure to follow documented training - Initial/regular training not provided or unknown to operator		
	Gloves		Hand protection	
		Hand protection as recommended and specified by the machine builder and/or machine owner. Typical residual risk: - Failure to properly use and maintain PPE Action to be taken		
	Operating manua		Instruction manual	
		Machine manual(s) to be available in accordance to the requirements of Machinery Directive 2006/42/EC. Machine manual(s) to be available in accordance to the local/regional requirements. Typical residual risk: - Failure to follow documented instruction - Documentation not readily available to operator - Documentation not in appropriate language(s)		
	Safety Shoes		Foot protection	
		Foot protection as recommended and specified by the machine builder and/or machine owner. Typical residual risk: - Failure to properly use and maintain PPE		
	Warning Signs		Awareness (safety) signs	

	Implemented safety awareness sign that complies to ISO 7010. To implement awareness sign as according to the requirements stated in ANSI B11.19. Typical residual risk: - Failure to heed warning - Failure to maintain visibility/legibility	
 Work procedures	 Safe work procedures	
	Safe work procedures to be developed in accordance to the requirements of Use of Work Equipment Directive 2009/104/EC. Safe work procedures to be developed in accordance to the local/regional requirements. Typical residual risk: - Failure to follow documented procedures - Procedure not provided or unknown to operator	

4.7 Risk list

4.7.1 R001

Mechanical hazards											
	Approach of a moving element to a fixed part										
Source			Consequence								
Tilting Movement Motor			Shearing								
-Shearing during tilting movement											
	Production Task										
Operator loads clutch plate using crane then places the clutch plate in the machine and starts the process operation with help of cycle start- two hand push buttons			Affected persons								
			Operator								
											
Severity [S]		Exposure [E]		Avoidance [A]		Occurrence [O]		Risk IN			
S2		E3		A2		O3		3			
Risk reduction							S	E	A	O	Risk OUT
Inherent design											
Engineering	Guards			Fix Guard	Fixed guard- Machine perimeter						
				Operator Access							
	Devices			Cycle Start	Light curtain						
				Start PB	Two-hand actuating control						
				Front Estop	E-Stop push button						
			Air isolation	Energy isolation pneumatic							
							S2	E1	A2	O3	<1

Administrative			Warning Signs						
			Gloves	Awareness (safety) signs					
			Safety Shoes	Hand protection					
			Operating	Foot protection					
			Instruction manual	Instruction manual					
			Safe work procedures	Safe work procedures					
Residual risk									

4.7.2 R002

Mechanical hazards											
		Clamping Cylinders									
Source			Consequence								
Pneumatic cylinders			Injection								
during the operation clamping cylinders are present in the machine to hold the component while tilting operation which helps the operator for installing the nuts with help of nut runner											
		Production Task									
Operator loads clutch plate using crane then places the clutch plate in the machine and starts the process operation with help of cycle start- two hand push buttons				Affected persons							
				Operator							
Severity [S]		Exposure [E]		Avoidance [A]		Occurrence [O]		Risk IN			
S2		E3		A2		O3		3			
Risk reduction							S	E	A	O	Risk OUT
Inherent design											
Engineering	Guards			Fix Guard	Fixed guard- Machine perimeter						
	Devices			Operator Access							
				Cycle Start	Light curtain		S2	E3	A1	O3	3
				PB	Two-hand actuating control						
				Front Estop	E-Stop push button						
				ON/Off Switch	Power Isolation switch						
			Air isolation	Energy isolation pneumatic							

Administrative			Warning Signs						
			Gloves	Awareness (safety) signs					
			Safety Shoes	Hand protection					
			Foot protection	Instruction manual					
			Operating manual	Safe work procedures					
			Work procedures	Training					
			A005						
Residual risk									

4.7.3 R003

Electrical hazards

	Overload					
Source		Consequence				
Electrical Power		Electrocution				
-if there is a sudden surge in the power possibilities of electrocution for the operator						
	Trouble Shooting					
Identify jam or misalignment Check sensors and actuators Manual repositioning which also covers Preventive Maintenance planned once in 15 days.		Affected persons				
		Maintenance person				
Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN		
S4	E3	A2	O3	10		

Risk reduction

					S	E	A	O	Risk OUT
Inherent design									
Engineering	Guards								
	Devices			ON/Off Switch	Power Isolation switch	S4	E0	A1	O1
Administrative				Operating manual	Instruction manual				
				Work procedures	Safe work procedures				
Residual risk									

4.7.4 R004

Mechanical hazards

	Approach of a moving element to a fixed part									
Source			Consequence							
Tilting Movement Motor			Shearing							
-Shearing during tilting movement										
	Cleaning/Maintenance									
Cleaning chips and oil			Affected persons							
Lubrication Cylinder or fixture maintenance										
			Operator							
Severity [S]		Exposure [E]		Avoidance [A]		Occurrence [O]		Risk IN		
S2		E3		A2		O3		3		
Risk reduction						S	E	A	O	Risk OUT
Inherent design										
Engineering	Guards									
	Devices	  	  	Operator Access Front Estop Air isolation	Light curtain E-Stop push button Energy isolation pneumatic	S2	E1	A2	O3	
Administrative		   	   	A005 Work procedures Operating manual Gloves	Training Safe work procedures Instruction manual Hand protection					
Residual risk										

4.7.5 R005

Electrical hazards

	Electrical Power					
Source	Consequence					
Short circuit in Control panel	Shock					
-						
	Trouble Shooting					
Identify jam or misalignment Check sensors and actuators Manual repositioning which also covers Preventive Maintenance planned once in 15 days.		Affected persons Maintenance person				
Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN		
S4	E3	A2	O3	10		

Risk reduction

					S	E	A	O	Risk OUT
Inherent design									
Engineering	Guards								
	Devices	 	ON/Off Switch	Power Isolation switch					
Administrative		   	 Work procedures  Operating manual  Awareness (safety) signs  Warning Signs A005	Safe work procedures Instruction manual Awareness (safety) signs Training	S4	E0	A1	O3	<1
Residual risk									

4.7.6 R006

Mechanical hazards

	Approach of a moving element to a fixed part			
Source		Consequence		
Tilting Movement Motor		Shearing		
-Shearing during tilting movement				
	Visitor			
Passing by machine Supervising Material movement nearby Persons being present in a hazardous area at or near the machine without a dedicated task at the machine / not working at the machine. This may include just walking by the machine, or observing the operation process without a defined/described associated task.		Affected persons		
		Visitor		
				
Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S2	E3	A2	O3	3

Risk reduction					S	E	A	O	Risk OUT
Inherent design									
Engineering	Guards			Fix Guard	Fixed guard- Machine perimeter				
	Devices			Operator Access	Light curtain				
Administrative				Warning Signs	Awareness (safety) signs				
				Safety Shoes	Foot protection				
Residual risk									

4.7.7 R007

Mechanical hazards											
		Clamping Cylinders									
Source					Consequence						
Pneumatic cylinders					Injection						
during the operation clamping cylinders are present in the machine to hold the component while tilting operation which helps the operator for installing the nuts with help of nut runner											
		Gripping Jaw setup									
Setting process changeover for 280 Model adjustment is done by changing the 3mm bolt and nuts at each stage the HMI has information for the operator for any sensor mis alignment during this process LOTO and administrative controls are followed.					Affected persons						
					Operator						
Severity [S]		Exposure [E]		Avoidance [A]		Occurrence [O]		Risk IN			
S2		E3		A2		O3		3			
Risk reduction							S	E	A	O	Risk OUT
Inherent design											
Engineering	Guards			Fix Guard	Fixed guard- Machine perimeter						
	Devices			Operator Access							
				Front Estop	Light curtain		S2	E3	A2	O3	3
				Air	E-Stop push button						
				isolatio	Energy isolation pneumatic						
				n	Power Isolation switch						
				ON/Off Switch	Two-hand actuating control						
				Cycle Start PB							

Administrative			Work procedu res	Safe work procedures					
			Operati ng	Instruction manual					
			manua Gloves	Hand protection					
Residual risk									

4.7.8 R008

Mechanical hazards													
		Clamping Cylinders											
Source					Consequence								
Pneumatic cylinders					Injection								
during the operation clamping cylinders are present in the machine to hold the component while tilting operation which helps the operator for installing the nuts with help of nut runner													
		Fixture base plate Change Over											
Setting for new base plate change over for 3 models - 230 , 280 , 310 three different base plates are added in the machine and the change over is done with a frequency (1hr to 1 week) depending on the production demand during this process LOTO and administrative controls are in place					Affected persons								
					Operator								
Severity [S]			Exposure [E]			Avoidance [A]		Occurrence [O]		Risk IN			
S2			E3			A2		O3		3			
Risk reduction									S	E	A	O	Risk OUT
Inherent design													
Engineering	Guards												
	Devices	    	ON/Off Switch Operator Access Front Estop	Power Isolation switch Light curtain E-Stop push button		S2	E1	A2	O3	<1			

Administrative			Gloves					
			Warning Signs	Hand protection				
			A005	Awareness (safety) signs				
			Work Training					
			Safe work procedures					
Operating manual				Instruction manual				
Residual risk								

5 Used symbols

5.1 Hazard symbols

Symbol	Hazard origin
	Overload
	Electrical Power
	Approach of a moving element to a fixed part
	Clamping Cylinders

5.2 Risk reduction measures symbols

	Discard measure
	Maintain measure
	Manipulation apparent
	Modify measure
	New recommended measure
	Replace existing device with similar device
	Replace existing measure with other measure

6 Impacts of legal requirements

6.1 Dealing with reporting information

- (1) The findings of the Machine Safeguarding Evaluation ADVANCED shall be provided in the form of a report. After completion of the work, the results shall be discussed at an agreed date with a responsible employee named by the customer.
- (2) The contents of this report, its interpretation and the implementation of recommended measures are addressed to persons whose technical qualification corresponds to the respective regional and/or local requirements for machinery safety.

6.2 Requirements from guidelines and regulations

If binding requirements exist for the design and/or execution of electrical, mechanical, hydraulic and pneumatic components from or on the basis of laws, directives, ordinances, standards or other sets of rules, which must be complied with in order to correspond to the state of the art or specifications of the customer, compliance with the respective specifications for the evaluated machine must be confirmed to SICK by the machine manufacturer or by the client.

6.3 Design Requirements

- (1) Hazards are considered according to standard ISO 12100, table B1 or according to a country-specific standard which is in reference to standard ISO 12100. For the services provided by SICK, it is assumed that existing components or plant components of electrical, mechanical, hydraulic or pneumatic nature are correctly selected and used and operated within their specifications.
- (2) Consequences that may arise from the fact that the inherent safety of a machine is not maintained, such as static and/or dynamic loading or overload, incorrect fastenings or assembly, insufficient dimensioning or incorrect calculation, or incorrect selection and design, are excluded from consideration.
- (3) During the examination of existing risk mitigation measures, the basic suitability/non-suitability for risk reduction shall be determined; it shall be assumed that such measures have been implemented and tested in accordance with their operating instructions and any applicable standards.

7 Risk assessment process

7.1 Risk assessment

The level of risk must first be determined and documented as the first step in determining safeguarding measures to reduce the risk of machine hazards. The methodology used by SICK to assess risk is in compliance with the processes described in ISO 12100 and ANSI B11 and is proven to establish a systematic approach to identify, evaluate and reduce risks for people as required by regulations for machine safety. This process must be documented as evidence of the information used to select the appropriate risk reduction measures and determine their minimum performance requirements, as appropriate.

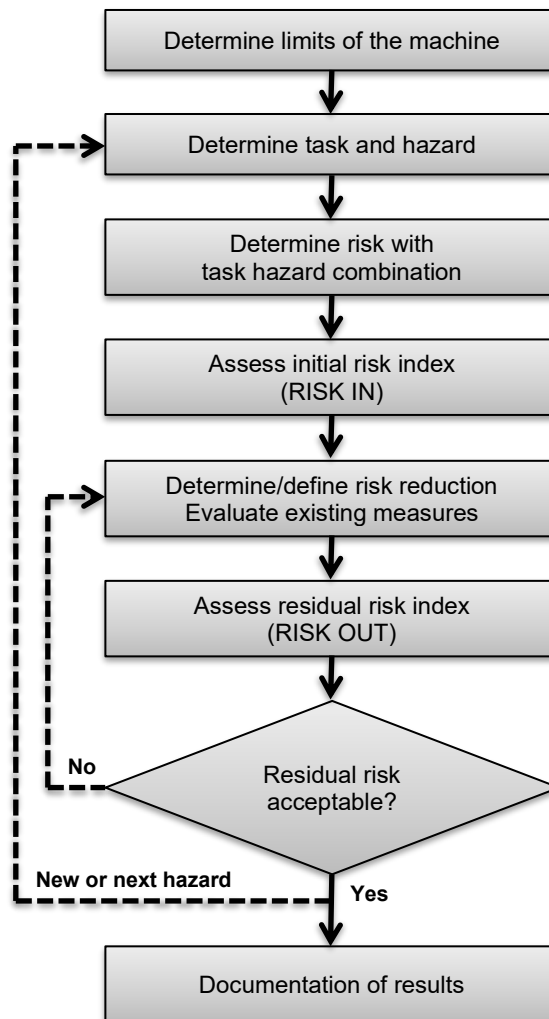


Figure 1: Overview of risk assessment process

7.2 Determination of the machine limits

7.2.1 Use Limits

Use limits include the intended use as well as the reasonably foreseeable misuse of the equipment. Aspects to be taken into account include the following:

- a) The different machine operating modes and the different intervention procedures for the users (including interventions required by malfunctions of the machine use),
- b) the use of the machinery (e.g., industrial, non-industrial and domestic) by persons identified by sex, age, dominant hand usage, or limiting physical abilities (e.g., visual or hearing impairment, size, strength) – if specific information is not available, the assessment should take into account general information about the intended user population (e.g., appropriate anthropometric data),
- c) the anticipated levels of training, experience, or ability of users, such as:
 - 1) Operators
 - 2) Maintenance personnel or technicians
 - 3) Trainees and apprentices
 - 4) The general public
- d) exposure of other individuals to the hazards associated with the machinery where it can be reasonably foreseen, including:
 - 1) Operators working in the vicinity, e.g., operators of adjacent machinery (i.e., persons likely to have good awareness of the specific hazards)
 - 2) Non-operator employees in the vicinity, e.g., administrative staff (i.e., persons with little awareness of specific hazards but likely to have good awareness of site safety procedures, authorized routes, etc.)
 - 3) Non-employees in the vicinity, e.g., visitors (i.e., persons likely to have very little awareness of the machine hazards or the site safety procedures; members of the public, including children where applicable)

7.2.2 Space Limits

Aspects to be taken into account include:

- a) Range of movement
- b) Space requirements for persons to interact with the machine, e.g., during operation and maintenance
- c) Human interaction, e.g., “operator-machine” interface
- d) “Machine-power supply” interface

7.2.3 Time Limits

Aspects to be taken into account include:

- a) The “life limit” of the machinery and/or of some of its components (e.g., tooling, parts that can wear, electromechanical components), taking into account its intended use and reasonably foreseeable misuse
- b) Recommended service intervals

7.3 Determination of tasks and hazards

Following the determination of the limits of the machinery, the essential step in any machine risk assessment is the systematic identification of reasonably foreseeable hazards, hazardous situations and/or hazardous events during all phases of the machine life cycle, i.e.:

- a) Transport, assembly and installation
- b) Commissioning
- c) Use
- d) De-commissioning, dismantling and disposal

It is assumed that, when present on machinery, a hazard will sooner or later lead to harm if measures are not taken to eliminate or provide protective measures. Only when hazards have been identified can steps be taken to eliminate them or reduce risks. To accomplish this hazard identification, it is necessary to identify the operations to be performed by the machinery and the tasks to be performed by persons who interact with it, taking into account the different parts, mechanisms or functions of the machine, the materials to be processed, if any, and the environment in which the machine can be used. Task identification should consider all those tasks associated with all the phases of the machine life cycle listed above as considered in the scope of work. Task identification should also take into account, but not be limited to, the following task categories:

- a) Setting
- b) Testing
- c) Teaching/programming
- d) Process/tool changeover
- e) Start-up
- f) All modes of operation
- g) Feeding machine
- h) Removal of product from machine
- i) Stopping the machine
- j) Stopping the machine in an emergency
- k) Recovery of operation from jam
- l) Re-start after unscheduled stop
- m) Faultfinding/troubleshooting (operator intervention)
- n) Cleaning and housekeeping
- o) Preventive maintenance
- p) Corrective maintenance

The aim is to identify all reasonably foreseeable hazards, hazardous situations or hazardous events.

7.4 Risk Estimation

The risk assessment methodology utilized by SICK is designed to easily estimate and then distinguish between high and low risk situations. This process is aimed at producing consistently reproducible and easily understandable results. In order to provide either rough or detailed estimations of the risk parameters, consideration of secondary and sometimes tertiary matrices can be utilized to adapt the process to the requirements and resources of each application. For each risk a risk estimation will be performed using the Scalable Risk Analysis and Evaluation Method [SCRAM] based on a primary evaluation of the risk parameters.

Severity	Exposure	Avoidance	Occurrence				PLr		
			O1 – O3	O1	O2	O3			
S1	E0	÷	< 1				a		
	E1 – E3	A1, A2		< 1	< 1	< 1			
S2	E0	÷	≤ 1				b		
	E1	A1		< 1	< 1	1			
		A2		< 1	1	1			
	E2	A1		1	2	2			
		A2		1	2	2			
	E3	A1		2	3	3			
		A2		2	3	3			
	S3	E0		÷	1				d
E1		A1			3	4	4		
		A2	3		4	4			
E2		A1	4		5	5			
		A2	5		5	5			
E3		A1	5		6	6			
		A2	6		6	6			
S4		E0	÷		1				
	E1	A1			6	7	7		
		A2		7	7	7			
	E2	A1		7	8	8			
		A2		8	8	8			
	E3	A1		8	9	9			
		A2		9	10	10			
				Risk Index					

Figure 2: Scalable Risk Analysis and Evaluation Method [SCRAM]

Parameter		Severity of harm S (simplified estimation)
S1	NEGLIGIBLE	None or negligible (trivial) injury (e.g., small bruises or superficial cuts) which either do not require any treatment or only treatment that is limited to simple and normally available first aid methods and equipment.
S2	SLIGHT	Injuries which can be treated with normally available first aid equipment but require the help of medically trained personnel. - OR - The injury (medical condition) will be reversed within three months without treatment, but under monitoring of a medical practitioner.
S3	SERIOUS	Injuries which require treatment by a medical practitioner but do not lead to permanent impairment. - OR - Injuries which lead to the loss or permanent damage of parts of the human body (but not total loss) with reversible medical condition.
S4	SEVERE	Injuries which lead to the death of one or more persons. - OR - Injuries which require treatment by a medical practitioner in a hospital and may lead to a permanent impairment or loss of parts of the body, limbs, or senses/abilities.

Parameter		Exposure to hazard E (simplified estimation)
E0	PREVENTED	Foreseeable exposure or access to the hazard(s) is: eliminated / controlled / limited by inherently safe design measures, - OR - exposure is prevented by mechanical and/or control-related safeguarding equipment which is selected and implemented as appropriate for the application. The implemented functional safety performance of the related SRP/CS must meet or exceed the required functional safety performance ($PL \geq PLr$). E0 not available for initial estimation of risk
E1	LOW	LOW (E1) can always be assumed if: the exposure frequency is once per work shift, - AND - the exposure duration is less than 3 minutes.
E2	MEDIUM	Use level 2 for determination of exposure to hazard. For details see: Whitepaper "Risk Assessment and Risk Reduction for Machinery Part 3" .
E3	HIGH	HIGH (E3) can always be assumed, if: access to the hazard zone is required, - AND - the exposure frequency is at least twice per work shift, - AND - each time the exposure duration is equal to or higher than 1 minute.

Parameter		Possibility of avoidance A (simplified estimation)
A1	AVOIDABLE	There are certain conditions that allow the avoidance of harm (such as skilled workers, slow movements, infrequent intervention, low-complexity processes, no sudden or unexpected movements with high acceleration).
A2	NOT AVOIDABLE	The avoidance is nearly impossible due to the lack of indication or awareness of the hazardous situation (such as fast hazardous events, insufficient surrounding space for evasion, high complexity processes, and/or the effect of routine on hazard awareness).

Parameter		Probability of occurrence O (simplified estimation)
O1	LOW	LOW (O1) can always be assumed if: the system is not prone to trouble, - AND - the hazard frequency is less than 33% of the presence time.
O2	MEDIUM	Use level 2 for determination of probability of occurrence. For details see: Whitepaper "Risk Assessment and Risk Reduction for Machinery Part 3" .
O3	HIGH	HIGH (O3) can always be assumed if: the system is prone to trouble, - AND - the hazard frequency is equal to or higher than 66% of the presence time.

7.5 Risk Evaluation

The initial risk estimation [RISK IN] describes the risk without considering any measures in place. This risk index describes the level of the existing risk on a scale from <1 (low risk) to 10 (very critical risk). Following risk estimation, a risk evaluation must be carried out to determine if risk reduction is required. If risk reduction is determined to be necessary, appropriate protective measures shall be selected and applied, and the risk assessment process repeated. As part of this iterative process, the designer must check whether additional hazards are introduced, or other risks increased when new risk reduction measures are applied. If additional hazards do occur, they shall be added to the list of identified hazards and appropriate protective measures will be required to address them.

7.6 Selection of Risk Reduction Measures

When the evaluation process determines that additional risk reduction measures are necessary, the selection and application of protective measures should be a decision made according to the risk(s) associated with each application.

When selecting risk reduction measures, it is important to consider that different measures have different effects on the various risk factors. With this in mind, it is always prudent to apply more than one protective measure in order to have the greatest cumulative effect on the reduction of risk. Furthermore, some protective measures are more susceptible to failures or error/misuse than others.

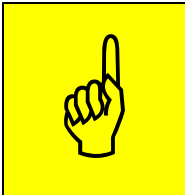
7.7 Evaluate Residual Risk

Following application of one or more risk reduction measures, the residual risk must again be assessed, this time accounting for the protective measures that are already in place as well as any new measures being added to the overall solution. In this report, the estimated residual risk is identified as RISK OUT and is based on the **assumption** that all recommended protective measures have been **implemented and validated according to applicable standards** and will be maintained over time through periodic inspection and maintenance.

The use of guards or safety functions very often results in people no longer being exposed to hazards or having no access. In this case, the parameter [E0] in the exposure group is used to estimate the residual risk. A hazard at which the probability of occurrence is prevented (E0) and additional administrative measures provided by the designer have been implemented by the user is rated with a residual risk index (RISK OUT) of AR (As low as reasonably practicable).

7.8 Achieving Acceptable (Tolerable) Residual Risk

It is important to understand that the intent of the risk assessment process is to identify risks which are not acceptable, apply protective measures based on the level of inherent risk present, and ultimately to reduce the risk to a level deemed acceptable or tolerable by both the organization as well as the affected individuals. While the aim is to achieve residual risk levels that are low or negligible, it is important to point out that the attainable level of residual risk cannot always achieve such a low degree. Instead, acceptable residual risk is often equated to the concept of 'As Low As Reasonably Practicable' (ALARP), which is a common best practice to judge the balance of risk and societal benefit.



The subjective nature of determining if acceptable risk has been achieved precludes SICK from stating whether the residual risk achievable is tolerable to any organization. As a partner with specialized experience and technical expertise, SICK can provide guidance and recommendations to your organization. In the end, however, acceptance (or rejection) of residual risk is the sole responsibility of the organization choosing to place or maintain industrial equipment in the workplace.

8 Information data protection

(Pursuant to Article 13 GDPR)

Subject	Description
Responsible body	SSU – place of jurisdiction SICK INDIA PRIVATE LIMITED 215, Western Edge II 400066 Mumbai
Purpose of data processing	The Machine Safeguarding Evaluation service supports the purpose to gain knowledge about the safety status of one or more machines in order to recommend appropriate protective measures in the following step
Categories of data	- Customer address - Photos/Sketches/Graphics - Writing down interviews
Legal basis	GDPR Art. 6 (1) f) – corporate interests
Corporate interests	Sick has the following interest: - Accurate / verifiable MSE - Storage for proof reasons (§199 Abs. 2 BGB) - Interest of the individual worthy of protection - No detailed report (anonymized)
Consequences of non-delivery of data	The provision of personal data is not required by law or contract. If the required data is refused by the customer, the quality of the evaluation decreases or the evaluation becomes impossible.
Data origin	30 years after preparation of the report
Data origin	SICK only processes data which have been provided to SICK directly or that are originated from the findings of the Machine Safeguarding Evaluation service.
Recipient of the data	SICK AG Erwin-Sick-Straße 1 79183 Waldkirch Germany
Transfers to third countries	No third countries
Logic of the automated decision	No automated decisions are made

You have the right

- for information about personal data concerning you;
- to correct personal data concerning you;
- to delete or block personal data concerning you;
- to object to the processing of personal data concerning you;
- the transferability of personal data concerning you;
- to revoke your consent with effect for the future, provided that any processing of personal data concerning you is based on your consent; and
- complain to a data protection supervisory authority if you believe that the processing of personal data concerning you violates data protection law.

9 Abbreviations

Information of used acronyms / customer definitions	
AOPD	Active opto-electronic protective device
AOPDDR	Active opto-electronic protective device responsive to diffuse reflection
ESPE	Electro sensitive protective device
PL	Performance level
PLr	Performance level required
PPE	Personal protective equipment

10 Signature

SICK Safety Specialist	
Name	
Title	

SICK Safety Specialist	
Name	
Title	

Customer Acknowledgement	
Name	
Title	

Customer Acknowledgement	
Name	
Title	

SICK AT A GLANCE

SICK is a leading manufacturer of intelligent sensors and sensor solutions for industrial applications. With more than 11,000 employees and over 50 subsidiaries and equity investments as well as numerous agencies worldwide, we are always close to our customers. A unique range of products and services creates the perfect basis for controlling processes securely and efficiently, protecting individuals from accidents and preventing damage to the environment.

We have extensive experience in various industries and understand their processes and requirements. With intelligent sensors, we can deliver exactly what our customers need. In application centers in Europe, Asia and North America, system solutions are tested and optimized in accordance with customer specifications. All this makes us a reliable supplier and development partner.

Comprehensive services round out our offering: SICK LifeTime Services provide support throughout the machine life cycle and ensure safety and productivity.

For us, that is "Sensor Intelligence."

Worldwide presence

Australia, Austria, Belgium, Brazil, Canada, Chile, China, Czech Republic, Denmark, Finland, France, Germany, Great Britain, Hungary, India, Israel, Italy, Japan, Malaysia, Mexico, Netherlands, New Zealand, Norway, Poland, Romania, Russia, Singapore, Slovakia, Slovenia, South Africa, South Korea, Spain, Sweden, Switzerland, Taiwan, Thailand, Turkey, United Arab Emirates, USA, Vietnam.

Detailed addresses and further locations: → www.sick.com